Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller

Software Architecture

Reference	TBD **[TBD-1]**
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Issue	Rev.	Date	Change
1	0	2026-10-11	First draft

1.0 Purpose and Scope

This document records the software architecture of the System: its components, what each does, how they connect, and the design decisions behind them [1, Sec. 4.2.3, p. 33]. It covers the three software items of the requirements [6, Sec. 3.1].

The architecture concentrates on "the most pressing risks" [3, Sec. 4]: the eight catastrophic and critical hazards that do not yet meet the fault tolerance rule [5, Sec. 5.0].

It names no hardware. It says what each component does, not which device does it, and ends with what the hardware must provide. The detailed design of each component, with its algorithms and data, is recorded before that component is coded [1, Sec. 4.3.2, p. 34].

How to read it. Section 3.0 shows every component on one figure. Section 6.0 holds the design decisions, each with the options weighed and the hazards it targets. Section 7.0 describes the components, and Appendix A shows which component meets each requirement. Section 9.0 lists what the hardware must provide.

2.0 Documents

[1] NASA, *NASA Software Engineering Requirements*, NPR 7150.2D.
[2] NASA, *NASA Software Engineering Handbook*, NASA-HDBK-2203, Version D, Topic 5.13 "SwDD – Software Design Description".
[3] NASA, *NASA Software Engineering Handbook*, NASA-HDBK-2203, Version D, SWE-057 "Software Architecture".
[4] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Concept of Operations*, Issue 1, Revision 1, Draft.
[5] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Hazard Analysis*, Issue 1, Revision 0, Draft.
[6] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Software Requirements Specification*, Issue 1, Revision 0, Draft.
[7] *Fail-Safe In-Orbit Reprogramming of a Spacecraft Controller — Software Classification*, Issue 1, Revision 1, Draft.
[8] NASA, *NASA Software Safety Guidebook*, NASA-GB-8719.13, 2004.

3.0 System Overview

The System has three software items [6, Sec. 3.1], each made of components (Figure 1). The ground software prepares and signs what the operator decides, and keeps what comes back. The onboard computer software checks everything that arrives, holds it, passes it on, and watches and restores the controller. The controller firmware runs the controller: a start-up selector chooses which firmware copy runs, the main firmware does the work, and the last-resort firmware takes over when the main firmware cannot.

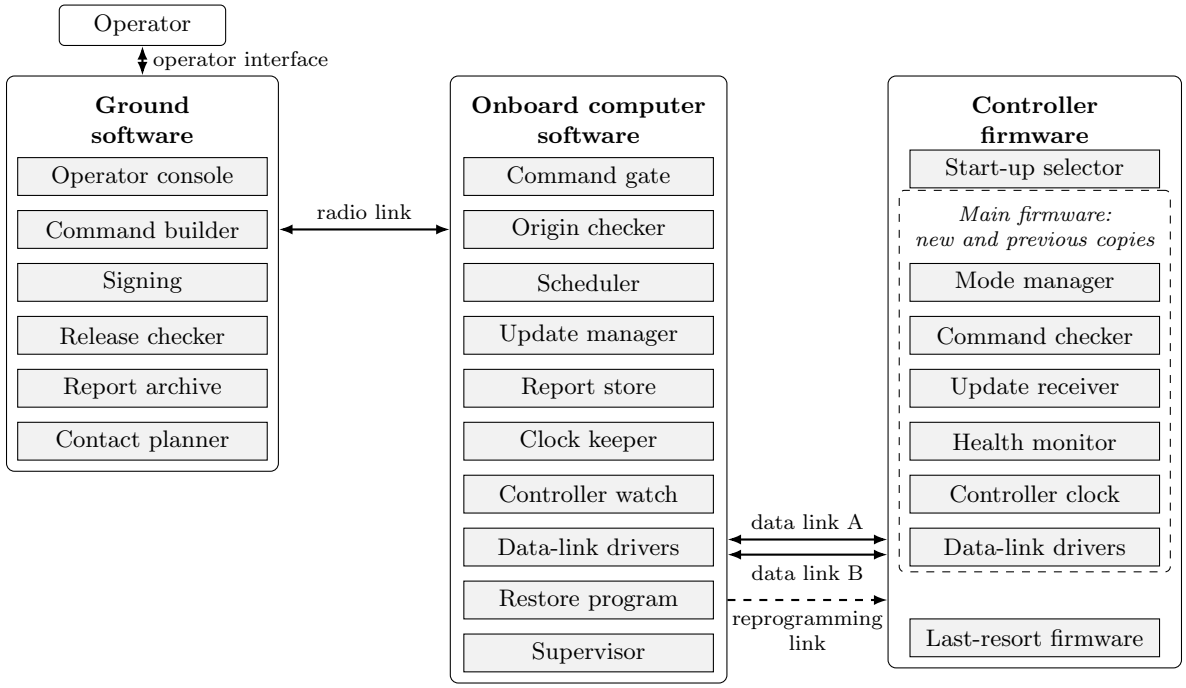


Figure 1: The three software items and their components. The data link has two connections, A and B. The dashed arrow is the reprogramming link, which reaches the controller whatever its memory holds.

4.0 Constraints

The architecture works within the constraints and assumptions of the ConOps [4, Sec. 1.1.2] and its accepted risks [4, Sec. 8.0]. Those below shape it.

C-1 Every command, report and firmware image passes through the onboard computer, so the onboard computer is on every path between the ground and the controller.

C-3 One engineer builds the System. Independence between components comes from separate code, not from separate teams.

A-2 Power can be lost at any moment, including while memory is being written.

A-5 Bit errors can change stored data and held commands.

A-6 The onboard computer's own software, including its operating system, is corrected by a system outside this project.

R-4 There is one onboard computer. Its permanent loss stops commands, reports and restores. The risk is accepted, so H-4 cannot meet the fault tolerance rule [5, Sec. 5.2].

C-4 The System runs on two given development boards. The design does not depend on them; they are checked against it at the end of this document.

5.0 Software Qualities

The software qualities [1, Sec. 4.2.3, p. 33] that drive this design are the two a high-risk project emphasises, fault tolerance and security [3, Sec. 4], and the testability the requirements demand.

Fault tolerance Every catastrophic hazard needs at least three failures together to occur, and every critical hazard two [5, Sec. 3.3].

Security Only the operator's commands and firmware are acted on (O-2).

Testability Each safety-critical component can be tested on its own on the bench, including under injected faults, to full modified condition/decision coverage [6, REQ-108].

6.0 Design Decisions

Each decision targets hazards that do not yet meet the fault tolerance rule [5, Sec. 5.2]. It states the problem, weighs the options, and lists the requirements the choice adds; these enter the requirements at their next revision.

Options are compared by the smallest number of failures that, together, cause the hazard with that option in place. The counts leave out the permanent loss of the onboard computer, an accepted risk (R-4). They are design targets, which the hazard analysis confirms in its design phase [5, Sec. 1.0]. Chosen: YES or NO.

6.1 Start-Up and Fall-Back

Problem. Every firmware copy, including the one the restore puts back, shares the start-up code and the data kept over restarts. One defect there stops them all, and the restore brings it back [5, Sec. 5.2]. Targets H-1 and H-8.

	Option	H-1	H-8	Chosen
a	The main firmware runs first and chooses its copy; the restore puts back a stored copy of it	2	1	NO
b	The last-resort firmware runs first and chooses; the restore puts it back	2	1	NO
c	A start-up selector, sharing no code with either firmware, runs first and chooses; the restore puts the last-resort firmware in its place	3	2	YES

Choice. Option c [**TBR-1**]. A defect in the start-up selector, or one in both main firmware copies, no longer stops the controller: the last-resort firmware shares no code with them [6, REQ-009] and runs in their place.

Adds.

- The start-up selector shall share no code with the main firmware or the last-resort firmware.
- The restore shall put the last-resort firmware in the start-up selector's place, so that it runs first.
- The onboard computer software shall put back the start-up selector and the main firmware through the reprogramming link when the operator orders it.

6.2 The Way Back In

Problem. A restore that goes through any controller software fails whenever that software fails, which the requirements forbid [6, REQ-056]. Targets H-1, H-3 and H-4.

	Option	H-1	Chosen
a	The restore is a command that the controller's firmware carries out	2	NO
b	The reprogramming link loads the controller's memory whatever it holds; the onboard computer drives it and can restart the controller	3	YES
c	Two such links, independent of each other	4	NO

Choice. Option b **[TBR-2]**. One link meets the rule; a second adds a connection to build and to check between uses [6, REQ-059] for a margin the rule does not ask for.

Adds.

- The reprogramming link shall load the controller's memory whatever that memory holds.
- The onboard computer software shall be able to restart the controller.
- No controller firmware shall be able to disable the reprogramming link.

6.3 Restore Apart From Update

Problem. The update path and the restore share onboard computer software and stored firmware. One fault there stops both until the outside system corrects it (A-6) [5, Sec. 5.2]. Targets H-3.

	Option	H-3	Chosen
a	One program updates and restores, from one firmware store	2	NO
b	A restore program, separate from the update manager, with its own firmware store	3	YES

Choice. Option b **[TBR-3]**. New firmware can still reach the controller through the restore program, which loads firmware from the ground [6, REQ-058], when the update path fails.

Adds.

- The restore program shall share no code or stored data with the update manager.

6.4 Proof of Origin

Problem. One means proves that both commands and firmware come from the operator. If it is disclosed, another party can command the controller and load firmware [5, Sec. 5.2]. Targets H-2 and H-5.

	Option	H-2	H-5	Chosen
a	One secret, held on the ground and on board, checks commands and firmware	1	1	NO
b	Digital signatures made with one key: on board is only the means to check them, so the key can be disclosed only on the ground	1	1	NO
c	Signatures made with two keys kept in two separate places; the radio link's own protection, outside this project, as a third barrier	2	2	NO
d	Signatures made with three keys kept in three separate places; every command and firmware image carries all three	3	3	YES

Choice. Option d **[TBR-4]**. Each disclosure exposes one key, so another party needs three. The controller checks firmware signatures again before running it [6, REQ-006], with code different from the onboard computer's, so that one defect in that code cannot pass forged firmware through both checks [8, Sec. 7.4.2, p. 141].

Adds.

- Every command and firmware image shall carry signatures made with three keys kept in three separate places on the ground.
- The onboard computer software shall act on a command or firmware image only when all three signatures are valid.
- The controller firmware shall check firmware signatures with code different from the onboard computer software's.

6.5 Two Clocks

Problem. Expiry, arm time-outs, scheduled commands, time-tags and the trial time limit all depend on the onboard clock. A wrong clock alone runs a scheduled command at the wrong time and gives reports wrong times [5, Sec. 5.2]. Targets H-6 and H-9.

	Option	H-6	H-9	Chosen
a	The onboard clock alone, compared with ground time at each contact [6, REQ-072]	1	1	NO
b	Counts instead of times wherever possible, such as report and command numbers	1	1	NO
c	A second clock in the controller, with its own time source, set only from ground time; each processor makes every time check with its own clock, and every report carries both times	2	2	YES

Choice. Option c [**TBR-5**]. A wrong clock on one processor is caught by the other. The controller's clock is never set from the onboard clock, so one wrong value cannot reach both.

Adds.

- The controller firmware shall keep its own clock, from a time source separate from the onboard computer's.
- The controller's clock shall be set only from ground time.
- The controller firmware shall refuse a command past its expiry, by its own clock.
- The controller firmware shall run a scheduled command only when its own clock agrees with the scheduled time within [**TBD-2**].
- Every report shall carry the time of both clocks.

6.6 Held Data

Problem. A bit error can change a held command, the arm state or the kept data after its check and before its use (A-5) [5, Sec. 5.2]. Targets H-6.

	Option	H-6	Chosen
a	Each item is checked once, when it arrives	1	NO
b	Each held item is stored with a check value and checked again just before use; the arm state is stored as a value that no single bit error turns into armed	2	YES
c	Memory that corrects bit errors on its own	2	NO

Choice. Option b [**TBR-6**]. A bit error must also produce a matching check value, a second failure. Unlike option c, it needs no hardware feature; option c can be added where the hardware has it.

Adds.

- Every held command and the kept data shall be stored with a check value and checked just before use.
- The arm state shall be stored so that no single bit error makes it armed.

6.7 Two Kinds of Connection

Problem. The two data-link connections share software at both ends, so one defect cuts both (R-1) [5, Sec. 5.2]. Targets H-4.

	Option	Failures to cut both	Chosen
a	Two connections of one kind, one driver at each end	1	NO
b	Two connections of one kind, a separate driver for each	1	NO
c	Two different kinds of connection, a separate driver for each	2	YES

Choice. Option c [**TBR-7**]. One defect, or one fault common to a kind of connection, now cuts one connection; the other carries on (O-3). H-4 still falls to the loss of the onboard computer (R-4).

Adds.

- The two data-link connections shall be of different kinds.

7.0 Components

For each component: its role, whether it is new or reused, and whether it is safety-critical [2, Sec. 3.3]. Origin: NEW, written for this System; MIXED, new code around a published library. Safety-critical: YES or NO, following the designation of its software item [7, Sec. 4.0].

7.1 Controller Firmware

Component	Role	Origin	Safety-critical
Start-up selector	Runs first at every start. Sets a safe state, checks each firmware copy for damage and origin, counts failed starts, and starts the new, previous or last-resort firmware.	MIXED	YES
Mode manager	Holds the controller mode, runs the trial, and confirms new firmware or gives it up.	NEW	YES
Command checker	Checks every command for mode, values, arming, expiry and repeats; checks held commands again before running them; answers and logs.	NEW	YES
Update receiver	Stores new firmware in the free copy, never over the previous or last-resort firmware.	NEW	YES
Health monitor	Tests the controller, detects faults, restarts it when it stops responding, and sends health and event reports.	NEW	YES
Controller clock	Keeps the controller's own time, set only from ground time.	NEW	YES
Data-link drivers	One driver for each data-link connection, sharing no code.	NEW	YES
Last-resort firmware	A separate program sharing no code or kept data with the others. Sets a safe state, answers commands, sends reports, and receives new firmware.	NEW	YES

The components from the mode manager to the data-link drivers form the main firmware, of which the controller keeps two copies: new and previous. The table below shows what runs in each controller mode [4, Sec. 3.4]; the start-up selector runs at every start, before any of them.

Mode	What runs
OPERATE	The main firmware, confirmed copy
TRIAL	The main firmware, new copy
FALLBACK	The main firmware, previous copy; or the last-resort firmware
RECOVERY	Nothing on the controller; the restore program works through the reprogramming link

7.2 Onboard Computer Software

Component	Role	Origin	Safety-critical
Command gate	Checks every command for origin, newness, expiry and arming, passes it on, and logs it.	NEW	YES
Origin checker	Checks the signatures on commands and firmware, and replaces the means of checking them on command.	MIXED	YES
Scheduler	Holds scheduled commands and passes each on once, at its run time, across restarts.	NEW	YES
Update manager	Checks new firmware, stores it, and delivers it over the data link, resuming after any interruption.	NEW	YES
Report store	Checks, numbers and time-tags every report, stores reports between contacts, and sends them at the next one.	NEW	YES
Clock keeper	Compares the onboard clock with ground time and applies corrections.	NEW	YES
Controller watch	Notices a silent controller, restarts it, and calls the restore when no firmware runs.	NEW	YES
Data-link drivers	One driver for each data-link connection, sharing no code.	NEW	YES
Restore program	A separate program with its own firmware store. Restores the controller through the reprogramming link, loads firmware from the ground through it, and checks its store and the link between uses.	NEW	YES
Supervisor	Restarts any program of the onboard computer software that crashes.	NEW	YES

7.3 Ground Software

Component	Role	Origin	Safety-critical
Operator console	Lets only the operator in, and shows reports, unanswered commands and the running firmware.	NEW	YES
Command builder	Builds every command from the shared command list and checks it before it is sent.	NEW	YES
Signing	Signs every command and firmware image with keys kept in separate places, and keeps a recoverable copy of the means of proving origin.	MIXED	YES
Release checker	Checks every firmware release before it is sent.	NEW	YES
Report archive	Keeps every report received and checks it for damage.	NEW	YES
Contact planner	Plans commands, uploads and report retrieval for coming contacts.	NEW	NO

The contact planner is the only component that is not safety-critical [**TBR-8**]. It passes plans to the command builder, which checks every command it builds.

8.0 Interfaces

Each interface is specified byte by byte before any code is written. Interfaces between components of one software item are set in its detailed design.

Interface	Between	Carries
Operator interface	Operator and ground software	The operator's decisions; reports and the System's state
Radio link	Ground software and onboard computer software	Signed commands and firmware; answers and reports
Data link A	Onboard computer and controller	Commands, answers, reports and firmware
Data link B	Onboard computer and controller	The same as data link A, over a different kind of connection
Reprogramming link	Restore program and controller	The controller's restart, and its memory contents

9.0 Hardware Needs

The hardware must provide what the decisions above assume. Each need is checked against the two boards before this document is released **[TBD-3]**.

Need	Needed by
A loading path into the controller that works whatever its memory holds, driven by the onboard computer	Section 6.2
A way for the onboard computer to restart the controller	Section 6.2
No way for controller firmware to disable that loading path	Section 6.2
A way for the restore to make the last-resort firmware run first	Section 6.1
Room in the controller for the start-up selector, two main firmware copies, the last-resort firmware and the kept data	Section 6.1
Kept data in the controller that survives a restart and a power loss	REQ-017
Room on the onboard computer for two separate firmware stores	Section 6.3
Processing in the controller to check three signatures within the restart time	Section 6.4; REQ-023
A time source in the controller separate from the onboard computer's	Section 6.5
Two different kinds of connection between the onboard computer and the controller	Section 6.7

A Requirement Allocation

Every requirement [6, Secs. 4.0–7.0] and the component that meets it. Requirements that the decisions in Section 6.0 add are allocated when they enter the requirements.

Component	Requirements
Controller firmware	
Start-up selector	REQ-004, REQ-005, REQ-006, REQ-013, REQ-014, REQ-015, REQ-016, REQ-017, REQ-018, REQ-019, REQ-023, REQ-113
Mode manager	REQ-001, REQ-002, REQ-003, REQ-011, REQ-012, REQ-013, REQ-020, REQ-021, REQ-024, REQ-025, REQ-026, REQ-113
Command checker	REQ-027, REQ-028, REQ-029, REQ-030, REQ-031, REQ-032, REQ-033, REQ-034, REQ-035, REQ-036, REQ-100
Update receiver	REQ-007, REQ-008
Health monitor	REQ-022, REQ-037, REQ-038, REQ-039, REQ-040, REQ-041
Controller clock	REQ-030
Data-link drivers	REQ-042, REQ-099
Last-resort firmware	REQ-004, REQ-009, REQ-010
Onboard computer software	
Command gate	REQ-064, REQ-065, REQ-066, REQ-067, REQ-087, REQ-100, REQ-112

Component	Requirements
Origin checker	REQ-043, REQ-064, REQ-084, REQ-085, REQ-086
Scheduler	REQ-068, REQ-069
Update manager	REQ-043, REQ-044, REQ-045, REQ-046, REQ-047, REQ-048, REQ-049, REQ-050, REQ-051, REQ-052, REQ-053, REQ-063
Report store	REQ-070, REQ-071, REQ-074, REQ-075, REQ-076, REQ-077, REQ-078, REQ-079
Clock keeper	REQ-072, REQ-073
Controller watch	REQ-061
Supervisor	REQ-062
Data-link drivers	REQ-080, REQ-081, REQ-082, REQ-099
Restore program	REQ-054, REQ-055, REQ-056, REQ-057, REQ-058, REQ-059, REQ-060, REQ-083
Ground software	
Operator console	REQ-091, REQ-094, REQ-095
Command builder	REQ-089, REQ-100
Signing	REQ-088, REQ-096
Release checker	REQ-090
Report archive	REQ-092, REQ-093
Contact planner	REQ-097
All software items	
Every component	REQ-098, REQ-101, REQ-102, REQ-103, REQ-104, REQ-105, REQ-106, REQ-107, REQ-108, REQ-109, REQ-110, REQ-111

B Open Items

The values below are not yet known.

TBD-1 Project name and document reference.

TBD-2 Largest allowed difference between the controller's clock and the time of a scheduled command.

TBD-3 Check of each hardware need against the two boards.

The proposals below await the author's acceptance.

TBR-1 Start-up and fall-back: option c (Section 6.1).

TBR-2 The way back in: option b (Section 6.2).

TBR-3 Restore apart from update: option b (Section 6.3).

TBR-4 Proof of origin: option d (Section 6.4).

TBR-5 Two clocks: option c (Section 6.5).

TBR-6 Held data: option b (Section 6.6).

TBR-7 Two kinds of connection: option c (Section 6.7).

TBR-8 The contact planner is not safety-critical (Section 7.3).